

Part 2: Implementation and Final Presentation

Objective

The objective of this part is to implement your data science project based on the plan you developed in Part 1. This includes data preprocessing, applying the chosen algorithms, visualizing the results, and presenting your findings.

Instructions

1. **Group Formation:** In “Project Part 2: Form a Group”, please choose the group. If you would like to stay in the same group, please choose the same group as project part 1. If for any reason, you would like to join a different group, you can choose a difference group.
2. **Data Preprocessing:**
 - Clean and preprocess your data using libraries such as NumPy.
 - Handle missing values, discretize the data, or convert text values to numerical values as necessary.
 - Document the preprocessing steps you took and justify your choices.
3. **Algorithm Implementation:**
 - Apply the chosen algorithms to your data using libraries such as Scikit-learn.
 - Ensure you understand the parameters and outputs of each algorithm.
 - Document the implementation process and any challenges you encountered. For example, "We used linear regression to predict accident rates based on traffic volume and K-means clustering to identify accident hotspots."
4. **Visualization:**
 - Create visualizations to support your analysis using libraries such as Matplotlib and Seaborn.

- Ensure your visualizations are clear, accurate, and effectively communicate your findings.
- Document the visualization techniques you used and justify your choices. For example, "We used scatter plots to show the relationship between traffic volume and accident rates, and heatmaps to visualize accident hotspots."

5. Presentation Preparation:

- Prepare a 10-minute presentation of your findings.
- Include an introduction, methodology, results, and conclusion.
- Use slides to organize your presentation and include visual aids.
- Practice your presentation to ensure it fits within the time limit. All group members should participate in the presentation.

Instructions for Presentation for Part 2

Presentation Structure:

1. Introduction:

- Briefly recap your hypothesis and chosen dataset. Discuss any potential ethics challenges
- Example: "Our hypothesis was that higher traffic volumes lead to increased accident rates. We used the Toronto Traffic Volume Data from the City of Toronto Open Data Portal."

2. Methodology:

- Describe the preprocessing steps you took.
- Explain the application of your chosen algorithms.
- Example: "We cleaned the data by removing outliers and used linear regression to analyze the relationship between traffic volume and accident rates. Additionally, we used K-means clustering to identify accident hotspots."

3. Results:

- Present your findings with visualizations.

- Discuss whether your hypothesis was supported.
- Example: "Our analysis showed a positive correlation between traffic volume and accident rates, as illustrated by our scatter plot. The clustering analysis identified several accident hotspots in downtown Toronto."

4. Discussion:

- Analyze the implications of your findings.
- Discuss any limitations and potential future work.
- Example: "While our findings support the hypothesis, further analysis with more data is needed to confirm the results. Future research could explore the impact of other factors such as weather conditions and road infrastructure on accident rates."

5. Conclusion:

- Summarize your key findings.
- Example: "In summary, our project demonstrated a significant relationship between traffic volume and accident rates, and identified key accident hotspots in Toronto."
- Open the floor for questions and feedback.

Project Part 2 Submissions

You should submit two files:

- Your presentation in the format of ".pdf".
 - Your presentation will be graded for CLO3-A and CLO3-B.
 - The 1 point for CLO3-B is graded individually. You will receive the point if you participated in the presentation. If you have accessibility needs, please contact the teaching team for alternative arrangements.
 - You can take a look at the grading scheme on eClass for CLO3-A (you can access it where you submit your work on eClass).
- Submit a text file (.txt) indicating whether each group member completed their assigned work. Free-riding is discouraged. If any member did not actively participate in the project (e.g., avoiding collaboration by working on other courses during lab time or being unresponsive), please report it in the file. We will follow up with the

reported member for clarification, and in serious cases, they may receive no points for any part of the project.

Please submit your files to “Project Part 2: Presentation”. You can submit your files as many times as you want.